Monaco Contouring — Quick Reference

Elekta Monaco TPS

Hot keys · navigation · contouring toolkit · Monaco gotchas · workflow tips

1 · Keyboard shortcuts (official)

Keys	All activities
`Ctrl` + `S`	Save patient data
`Ctrl` + `Z`	Undo last **contour** or block edit
`Ctrl` + `X`	Cut selected contour
`Ctrl` + `C`	Copy selected contour
`Ctrl` + `V`	Paste cut / copied contour
`PgUp` / `PgDn`	Next superior / inferior image (transverse)
`Num +` / `Num −`	Zoom in / zoom out (numeric keypad)
`F1`	Online help (searchable — try "shortcut")
`Ctrl` + `P`	Print BEV / DRR
`Ctrl` + `PrtScn`	Print entire screen
`Ctrl` + `Alt` + `PrtScn`	Active window → clipboard (teaching slides, M&M)
`Ctrl` + `Q` or `Alt` + `F4`	Exit Monaco

Keys	Contouring / editing
`Esc`	Abort an incomplete drawing / edit
`Delete`	Delete the **selected contour** (this slice) or port object
`Shift` + `Delete`	**Remove the whole structure** from the studyset
`→` `↑` `>`	Increase guide radius
`←` `↓` `<`	Decrease guide radius
Right-click in T/S/C/O view	Pop-up menu to pick a contouring tool without going to the ribbon

Three keys that bite

- `Ctrl` + `Z` only reverses **contour / block edits** — not margins, booleans or structure deletion. Save before big operations.
- `Delete` removes the **selected contour on that slice**; `Shift` + `Delete` removes the **entire structure**. Know which one your finger is on.
- `Esc` cancels a half-drawn contour — use it instead of clicking out and leaving a stray open segment.

2 · Navigation & image display

Navigation tool	Use it for
Mouse scroll wheel	Step through slices in the active view
Colour-coded T-bars	Drag to move the S/C/O plane displayed in the other views
Quick locator / Jump to point	Centre all views on a point — fast for checking a contour in 3 planes
Slice navigator / reference plane values	Type a slice position directly (useful for matching a diagnostic report)
Window / level by mouse	Drag in the image; or set from the Tools tab
W/L presets (user-definable)	Save your own (e.g. prostate, lung, mediastinum) — one click per switch
Link W/L affect	Apply one W/L to linked image sets — handy for planning CT + fused MR
Magnifying glass / zoom area	Zoom a region (apex, chiasm, cochlea) without losing context
Reset zoom & pan	Recover a lost image fast
Maximise / restore window	Contour on one large view, restore to check others
Layouts — save as preset	Store a contouring layout (e.g. big axial + small sag/cor) as a user default

Speed set-up (do once, saves minutes every case)

- Build personal **W/L presets** and a **saved layout**; set it as your user default.
- Learn the guide-radius arrows (`↑` / `↓`) so you never leave the image to resize the Replace tool.
- Use **right-click → tool** instead of travelling to the ribbon; keep one hand on the keyboard for `PgUp` / `PgDn` , `Esc` , `Ctrl` + `Z` .
- Turn on **Autosave Contours** so a crash doesn't cost you the session.
- Use **Anatomical Groups** to show/hide related structures (e.g. all H&N OARs) in one click.

3 · Contouring toolkit — what to reach for

Tool	What it does	Best use / pro tip
Draw Contour	Click points (or drag) to outline a structure on a slice.	Small or irregular targets; anything needing precise, deliberate edges (GTV).
Replace + Swap	Redraw a segment; Monaco splices it in. Swap flips to the other possible result if it kept the wrong side.	The workhorse edit. If the splice looks wrong, hit Swap before undoing.
Guide radius	Sets the snap distance for Replace / reshape edits.	Arrow keys or < > adjust on the fly. Small for fine edits near OARs; large for broad corrections.
Reshape	Push/pull an existing contour.	Smoothing bumps on a CTV after margin expansion.
Contour by Shapes	Drop circles / rectangles.	Quick blocking structures, ring aids, or palliative field references.
Paintbrush	Paint voxels with a user-set width; contour forms when you exit the tool. Has its own erase, smoothing and transparency.	Fast for soft-tissue OARs and editing CTVs. Paint with a slightly translucent fill to see anatomy beneath.
Edge Detection (Paintbrush)	Brush stops at image edges; sensitivity adjustable.	Works best on high-contrast boundaries (bladder–fat, lung–tissue); poor on iso-dense interfaces (prostate–rectum).
Structure Avoidance (Paintbrush)	Brush cannot paint into selected structures.	Paint a CTV while avoiding bone / bowel / air; paint rectum while avoiding prostate — no overlap to clean later.
EZ Sketch	Semi-automatic 3D structure creation, then Refine (local) or Global edit; Quick Scheme for repeat use.	Bladder, femoral heads, kidneys, liver. Always scroll every slice afterwards.
Auto Threshold	HU-based contour.	External/body, lungs, airway, bone. Set HU limits deliberately for lung window.
Auto Segmentation	Model-based auto-contours.	Starting point for OAR sets — review and edit, never accept unseen.
Interpolation	Fills slices between two drawn contours.	Contour every 2–3 slices on smooth OARs, interpolate, then check each interpolated slice (fails at bifurcations / apex).
Copy Sup/Inf, Ant/Post, R/L	Copies a contour to adjacent slices (from T, S or C views).	Tubular structures (spinal canal, rectum); then edit with Replace.
Copy Structure	Duplicate a whole structure.	Make an editable CTV copy before cropping; keep the original as an audit trail.
Auto Margin	Uniform / anisotropic 3D margins; save **Common Margin Templates**.	GTV→CTV, CTV→PTV, PRVs, rings. Save departmental templates once.
Advanced Margins	Boolean-style combinations (union, subtract, intersect) with margins; save templates; print margin report.	Crop PTV from skin, CTV minus bone, rectum minus PTV for optimisation, PTV_eval structures.
EZClean	Removes stray islands / tidies contours.	Run after paintbrush or threshold work to remove specks that inflate volumes.
PET SUV / threshold	SUV calculation and threshold contouring on fused PET.	Confirm SUV units first (DICOM editor for SUV if missing). Treat as a guide, not the GTV.
Generate Bolus	Creates bolus structure (incl. Fill VOI) with editing tools.	Chest wall / skin cases — confirm thickness & density with planning.
Convert Isodose → Structure	In Plan Review: turns an isodose into a structure.	Optimisation helpers, re-irradiation overlap, recording composite high-dose regions.

4 · Monaco-specific gotchas

Issue	What to do
Voxel-based structures	Monaco rasterises structures to voxels; a voxel is assigned to a structure once roughly half of it lies inside the contour. Thin or tiny structures (urethra, chiasm, cochlea, small mets) can lose or gain volume — check the reported volume and DVH, and contour generously-but-accurately rather than 1-voxel-thin.
Jagged external / body	Monaco re-voxelises the CT on import, so contours follow the voxel grid and the external can look stepped (relevant for SGRT surfaces). Applying a **0 mm Auto Margin** to the external is a commonly used way to smooth it.
Structure layering & overlaps	When structures overlap, the layering order in the IMRT constraints decides which owns each voxel during optimisation. Create explicit optimisation structures (PTV minus OAR, OAR minus PTV) rather than relying on layering alone.
Contours outside the patient	Structures extending beyond the external (e.g. PTV into air) need cropping or deliberate density handling — crop PTV 3–5 mm under skin for optimisation where your protocol allows.
Scan reference point	Define and **lock** it once contouring is complete; moving it late creates downstream set-up errors.
Structure mismatch at planning	Renaming or deleting structures after a template is loaded triggers mismatches — finalise names (TG-263 style) before the planner starts.

5 · Workflow tips & tricks

Efficient sequence for a typical case

- Fuse first (MR/PET) and check the registration locally around the target, not globally.
- External → auto-segment / EZ Sketch OARs → edit → targets last, when your eye is fresh.
- GTV on **every slice**; OARs every 2–3 slices + interpolation, then verify.
- CTV via Auto Margin from GTV, then edit against anatomical barriers using paintbrush with **structure avoidance**.
- PTV and PRVs **only** by Auto Margin / templates — never hand-drawn.
- Run EZClean, then a final scroll in sagittal and coronal: steps and gaps jump out.

Quality habits

- Change W/L for each structure: soft-tissue for pelvis, lung window for airways/lung, bone for canal and sacral plexus.
- Check superior/inferior extent on sagittal & coronal — the commonest miss in axial-only contouring.
- Use colour-wash / transparency to spot overlaps between CTV and OARs.
- Keep a _orig copy before major edits; it's your audit trail and undo for non-undoable steps.
- Consistent naming and colours via structure templates help peer review and auto-planning.
- Screen-grab with `Ctrl` + `Alt` + `PrtScn` for peer-review or teaching cases (de-identify first).

6 · Window / level starting points

Preset	Window	Level	Use for
Soft tissue / pelvis	400	40	Prostate, bladder, rectum, gynae CTVs
Mediastinum	350	50	Heart, oesophagus, great vessels, nodes
Lung	1500–1600	−600	Lung GTV, airways, lung OARs
Bone	1800–2000	400–500	Canal, vertebrae, femoral heads, sacrum
Brain	80	40	Brainstem, chiasm boundaries (pair with MR)
Liver	150	30–60	Liver lesions with contrast

Typical starting values — tune to your scanner, contrast phase and departmental protocol, then save as user-definable presets.

My department / version notes

Monaco version: _____ Local margin templates: _____

Extra shortcuts in my version (check F1): _____

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